

Machine Learning - Deep Learning fundamentals (69152)

Master in Robotics, Graphics and Computer Vision
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Faculty

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Office hours in Google Meet or in person

<https://directorio.unizar.es/?#/tutoria?colectivo=PDI&codCentro=110>

M - 12.00-14.00h

J - 10.00-14.00h

Please, send me an email to book a slot 24h in advance to confirm and specify if it's in person or online.

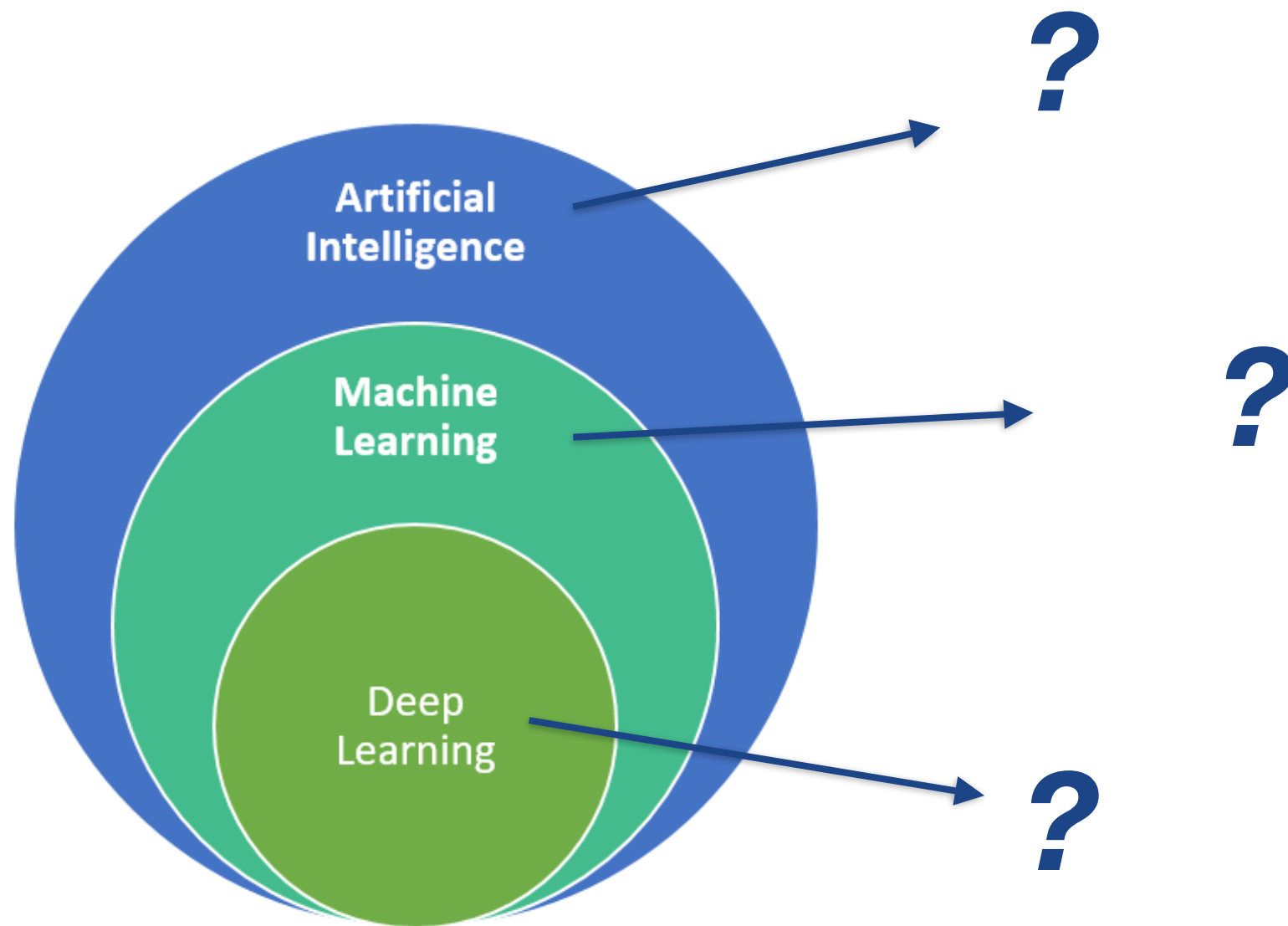
Tentative program for the DL block

Hours	Day	Topic
		...
M - 13-14h	2-Oct	Fundamentals of DL
Tu - 10-12h	3-Oct	Fundamentals of DL
M - 13-14h	16-Oct	CNNs
Tu - 10-12h	17-Oct	CNNs
LAB2	18-20Oct	<i>CNNs - Basics. Clasification. (Detection. Segmentation). Sample CV application.</i>
M - 13-14h	23-Oct	Finish CNNs
...	...	***** <i>Bayesian & RL BLOCKS</i> *****
Tu - 10-12h	21-Nov	<i>DRL</i>
		<i>Unsupervised</i>
LAB4	24 & 29 NOV	<i>Paper Reading lab</i>
		<i>Unsupervised. GANs</i>
		<i>RNNs, Transformers</i>
LAB5	13-15 Dec	<i>More CNNs. Unsupervised. DRL. (GANs). Sample Graphics and Robotics application.</i>
		<i>More recent DL</i>
		<i>More recent DL</i>

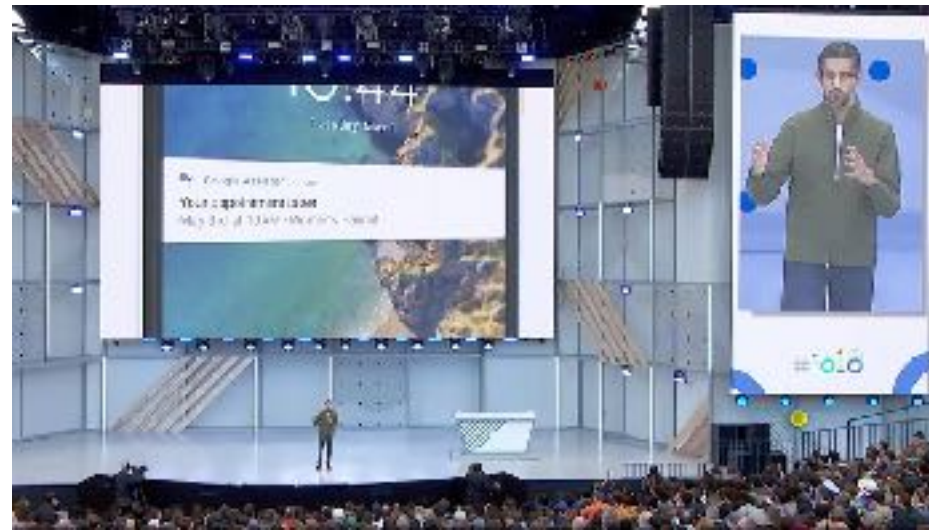
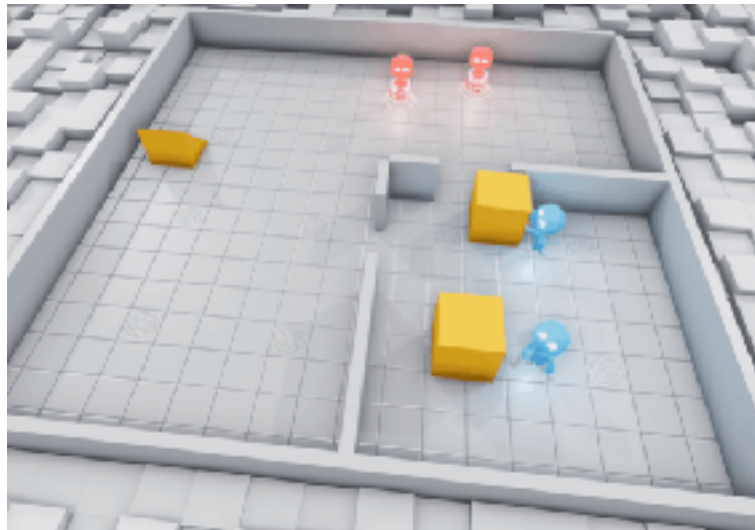
Today's class

- **What's deep learning?**
 - **Related topics**
 - **DL pipeline**
- Fundamentals of DL
 - Review basic concepts
 - NN and DNN

What's *deep learning*? (and what's not ...?)



What can we do with deep learning?



And what not ... :-)



What can we do with deep learning?

- Some examples ...

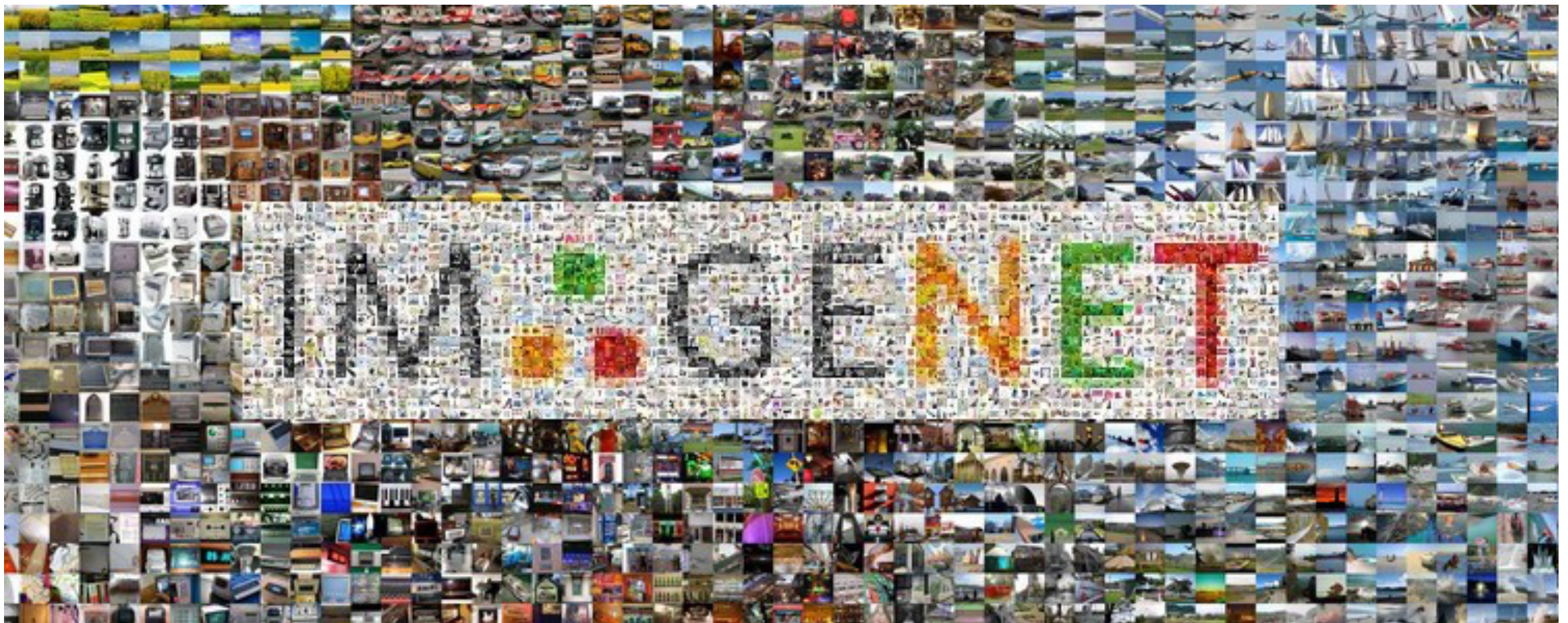
Some examples of successful DL applications ...

- **Object recognition**

Objects

- AlexNet. Competed in the ImageNet Large Scale Visual Recognition Challenge in 2012
- ImageNet: over 10 million annotated images

<http://image-net.org>

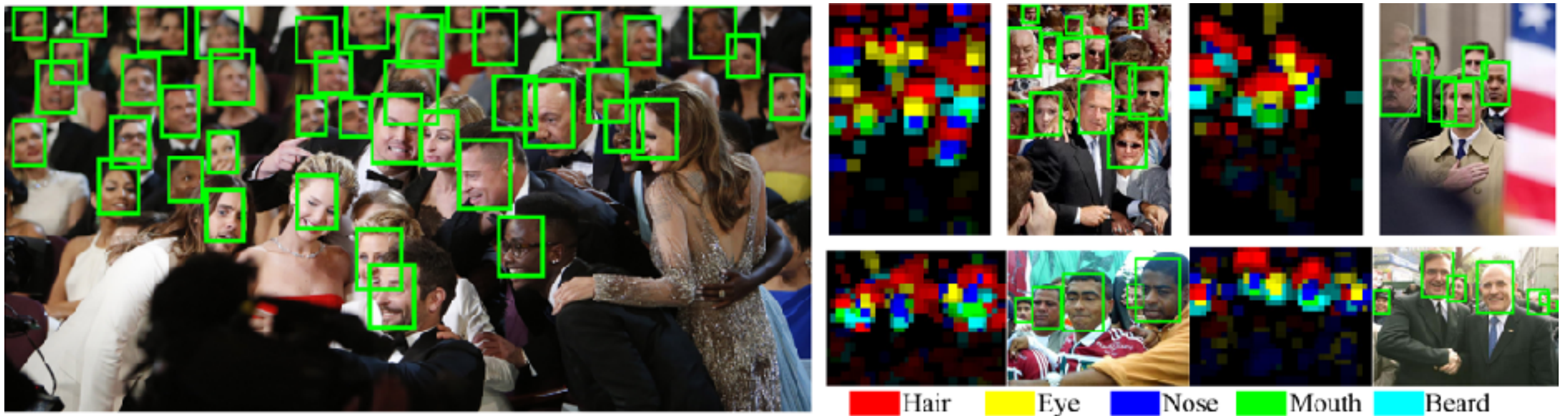


Some examples of successful DL applications ...

- **Face detection**

People

- CelebFaces Attributes Dataset (CelebA). Large-scale face attributes dataset, more than 200K celebrity images



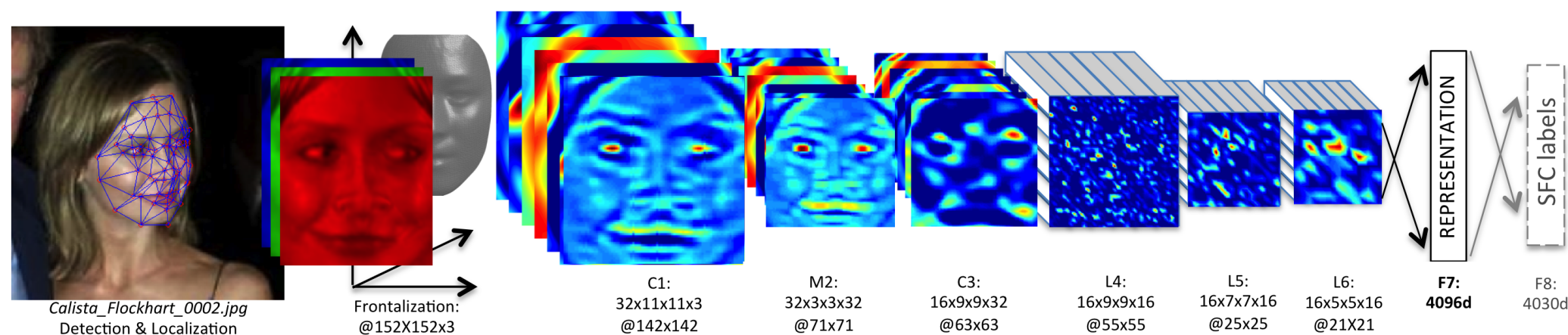
<https://www.youtube.com/watch?v=IMPjPQSb9g8&feature=youtu.be>

Some examples of successful DL applications ...

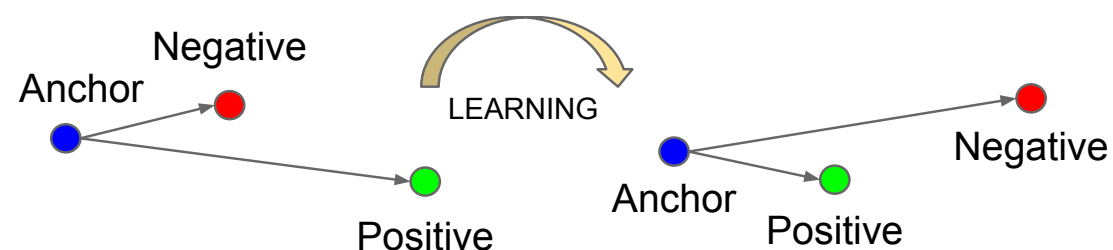
- **Face recognition**

People

- 4 million images of 4,000 persons



DeepFace: Closing the Gap to Human-Level Performance in Face Verification.
Yaniv Taigman Ming Yang Marc'Aurelio Ranzato Lior Wolf. CVPR 2014. Facebook



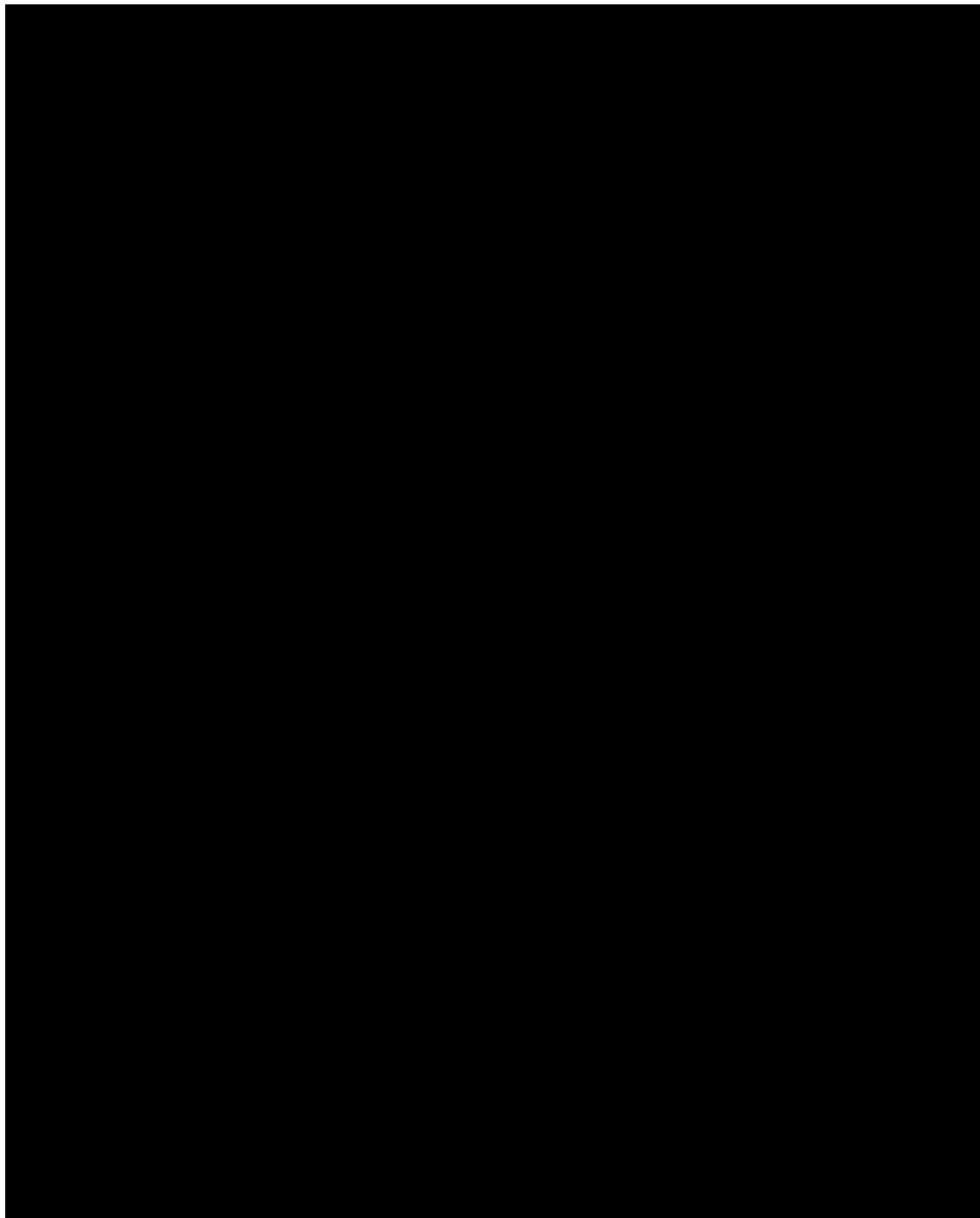
FaceNet: A Unified Embedding for Face Recognition and Clustering.
Florian Schroff, Dmitry Kalenichenko, James Philbin; CVPR 2015. Google



Some examples of successful DL applications ...

- Learn by watching how to play?! —> DRL

Behaviours

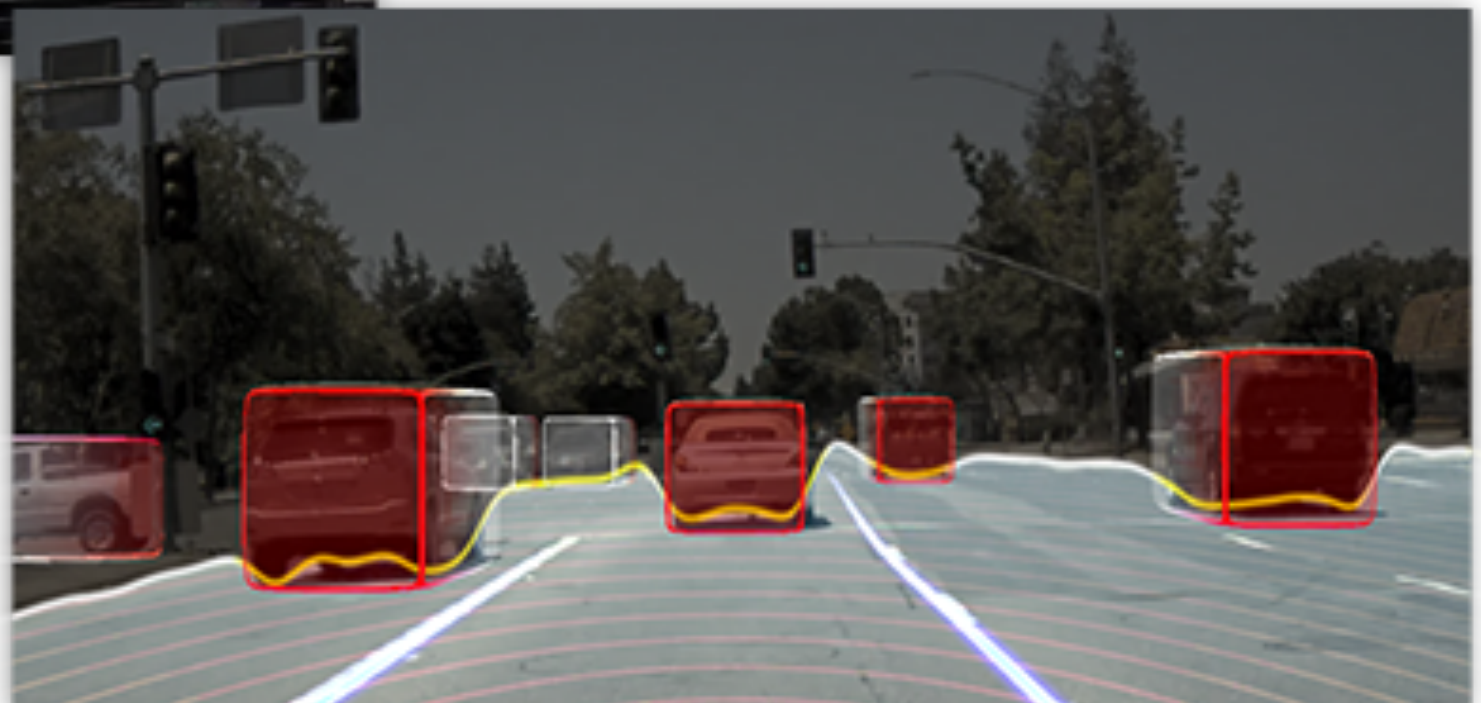


Google's Deep Mind

Some examples of successful DL applications ...

- Self-driving car

Behaviours



Some examples of successful DL applications ...

- DeepMind AI ***Not only visual tasks ...***
- Reduces Google Data Centre Cooling Bill by 40%
- *“Using a system of neural networks trained on different operating scenarios and parameters within our data centres, we created a more efficient and adaptive framework to understand data centre dynamics and optimize efficiency.”*

Google's Deep Mind

Some examples of successful DL applications ...

- **Voice** recognition and Natural **Language** Processing

Not only visual tasks ...



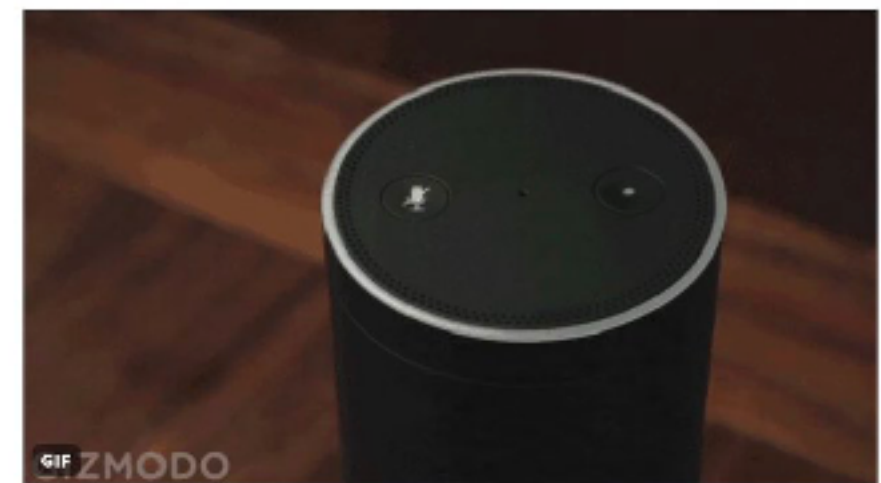
Hi, how can I help?



Jeff Bezos: "Alexa, buy me something from Whole Foods."

Alexa: "Sure, Jeff. Buying Whole Foods now."

Jeff Bezos: "WHA- ahh go ahead."



6/16/17, 9:23 AM

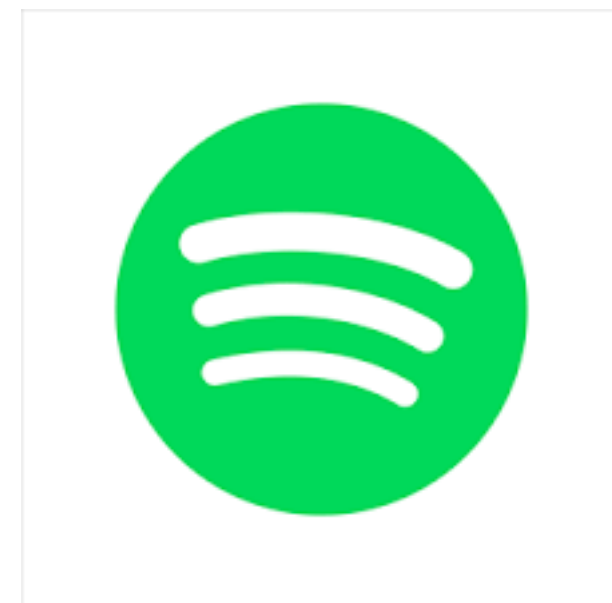
Tweet your reply



Some examples of successful DL applications ...

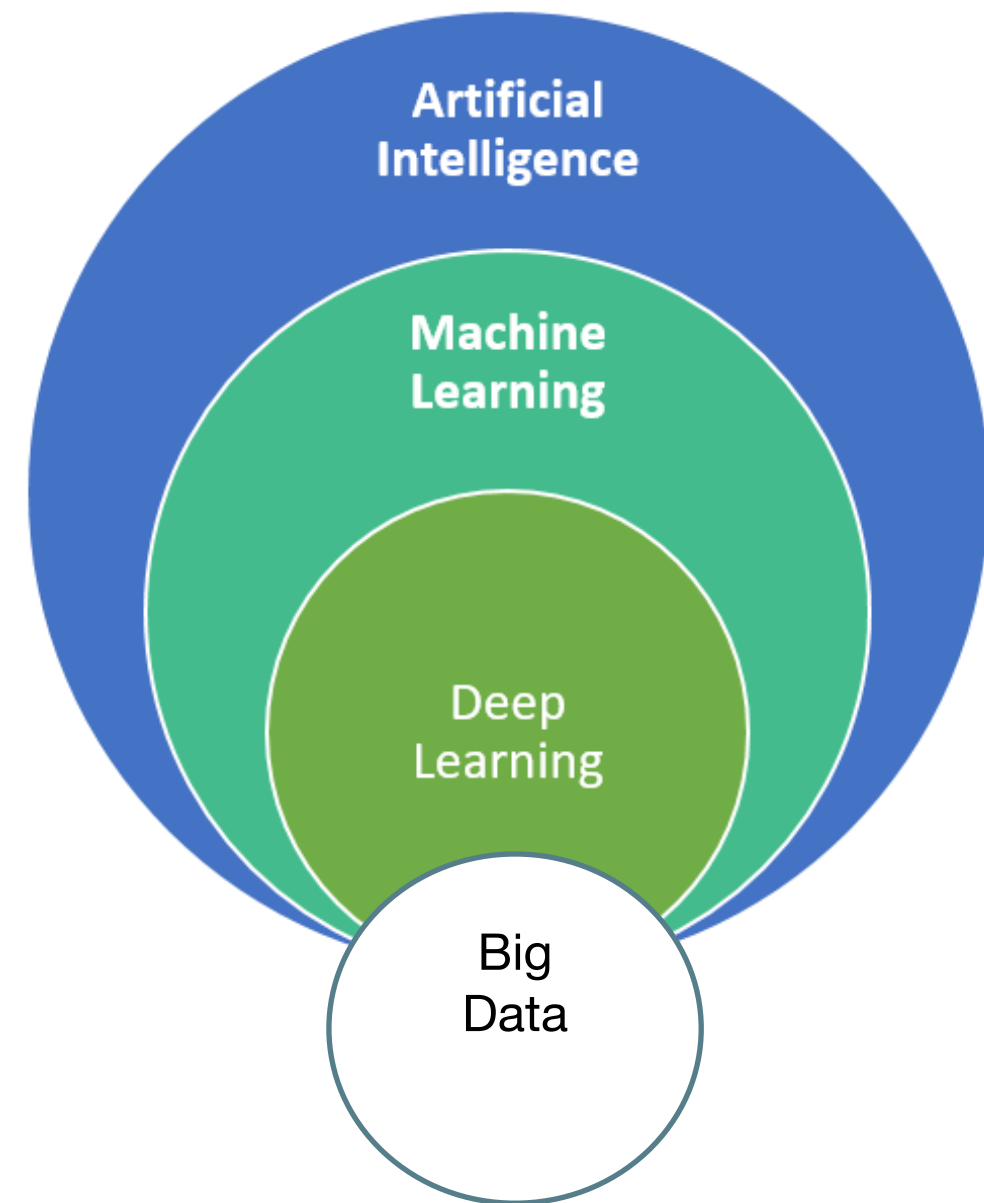
- **Recommendation** systems

*Not only visual
tasks ...*



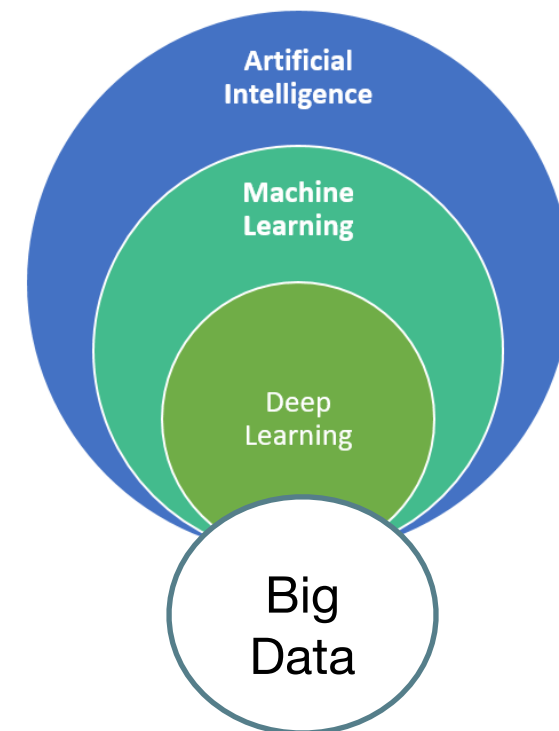
Deep Learning and Big Data

- Related topics to DL
 - Big data
 - Data science
 - Business Intelligence
 - ...



Deep Learning and Big Data

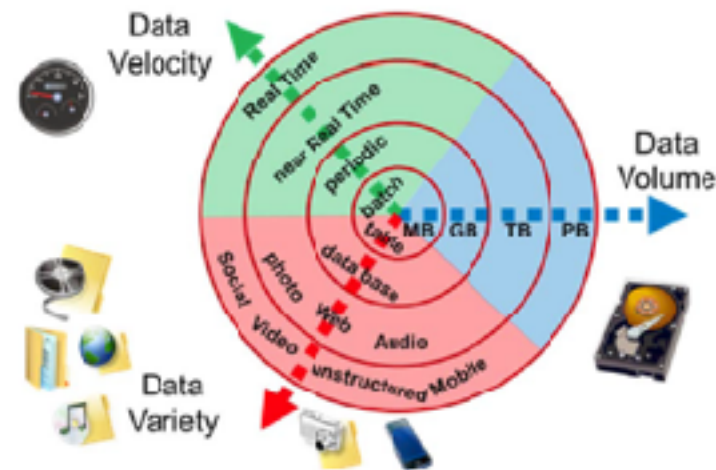
- Related topics to DL
 - **Big data:** *“data sets that are so large or complex that traditional data processing application software is inadequate to deal with them”* (Wikipedia)
 - **Data science:** Scientific field with the aim of extracting knowledge from raw data (usually “big data”)
 - **Business Intelligence:** Set of techniques to extract information from business data, that will be used to improve its performance.
 - *Data mining:* The exploration of raw data in the search for patterns (usually in the context of **databases**).



Machine Learning + Big Data



+



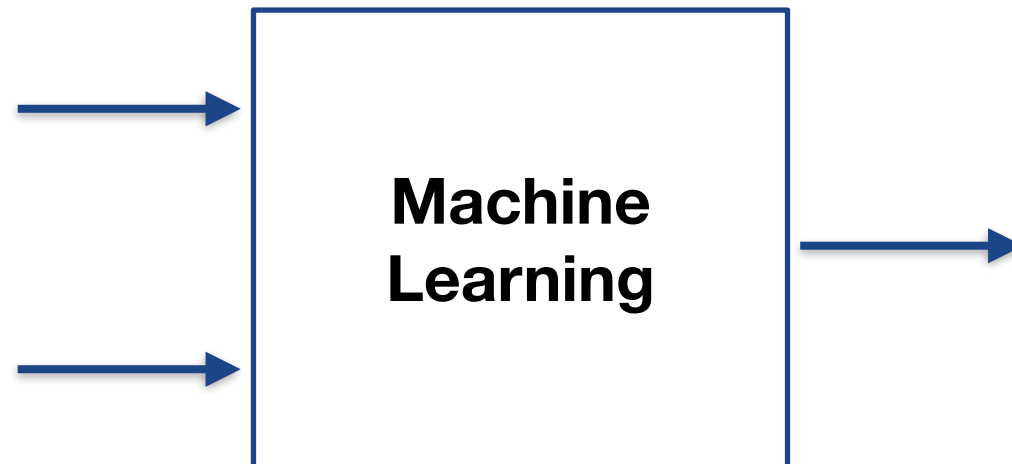
- **Cons:**

- *Torture the numbers, and they'll confess to anything — Gregg Easterbrook (and probably many others ...)*

- *So it's like having billions of monkeys typing. One of them will write Shakespeare. — Michael Jordan*

- **Pros:** New opportunities, challenges, and amazing results and applications!!

Deep Learning vs general ML pipeline



Deep Learning vs general ML pipeline

- Same methodology you saw the first day. Certain steps/ideas get particularly important: data selection, avoid overfitting, ...

1. Frame problem

2. Get data

3. Explore data

4. Prepare data

*It may sound you just
“throw” as much data as
you can ... but not just that!*

5. Explore promising models

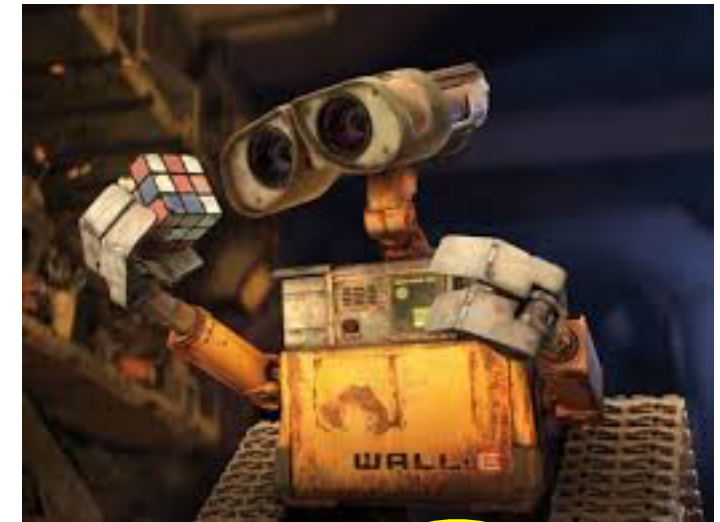
expensive!

6. Fine tune the best

7. Present/**analyze** the solution and results

ML pipeline & DL

- Collect **data** (labeled?) choose how to **represent** it?
- **Learn**: organize the sample data (training data): fit/learn a **model**
- **Classify**: **evaluate** new data with regard to the model learnt (classify/recognize new content)



How to get training data?

Lots of strategies. Crowdsourcing is a common ingredient lately

- ESP game. Luis Von Ahn and Laura Dabbish 2004
- LabelMe (MIT) - researchers and volunteers Russell, Torralba, Freeman, 2005
- Birds-200-2011 - fans of certain topic. Caltech-UCSD. Wah, Branson, Welinder, Perona, Belongie, 2011

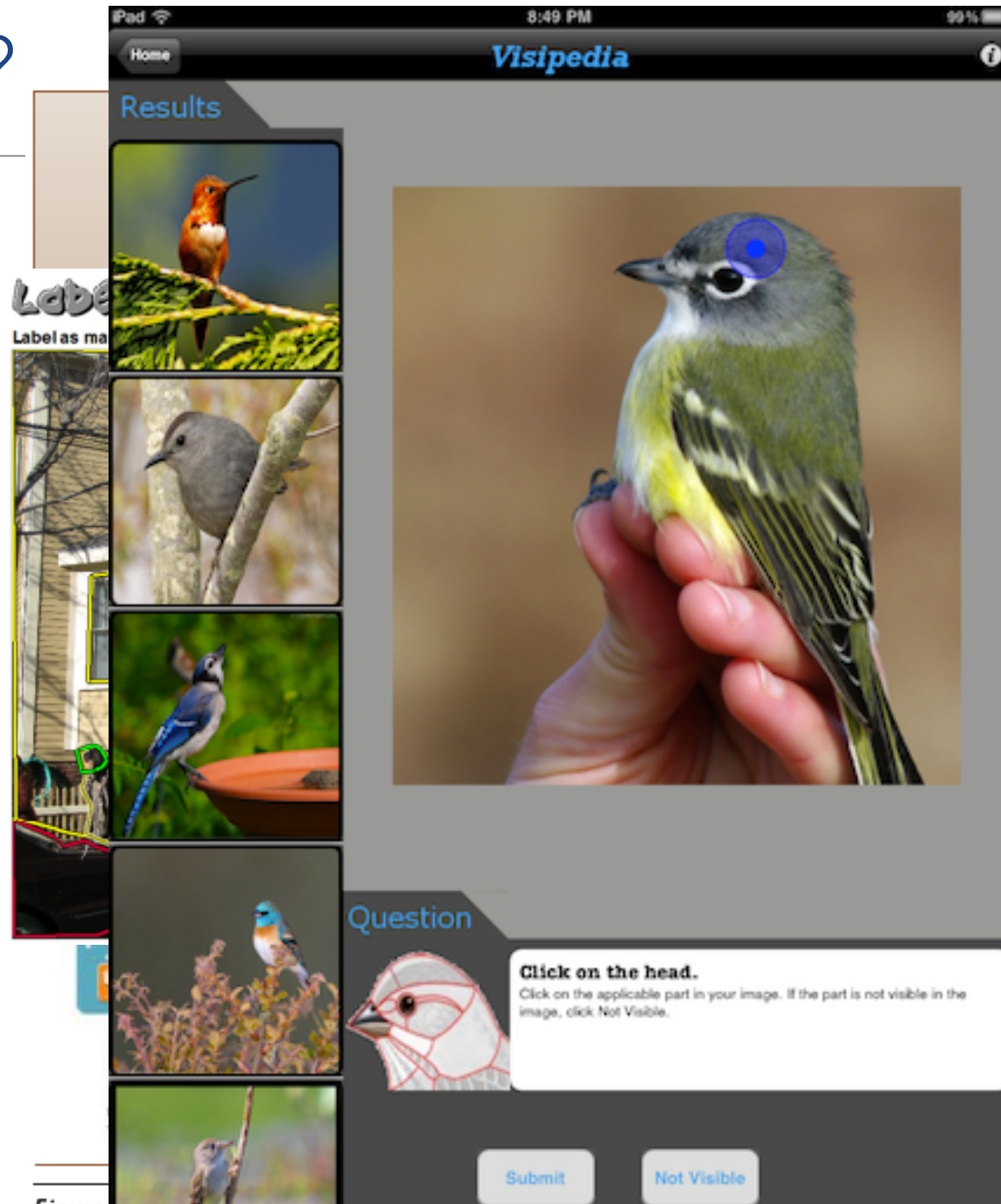
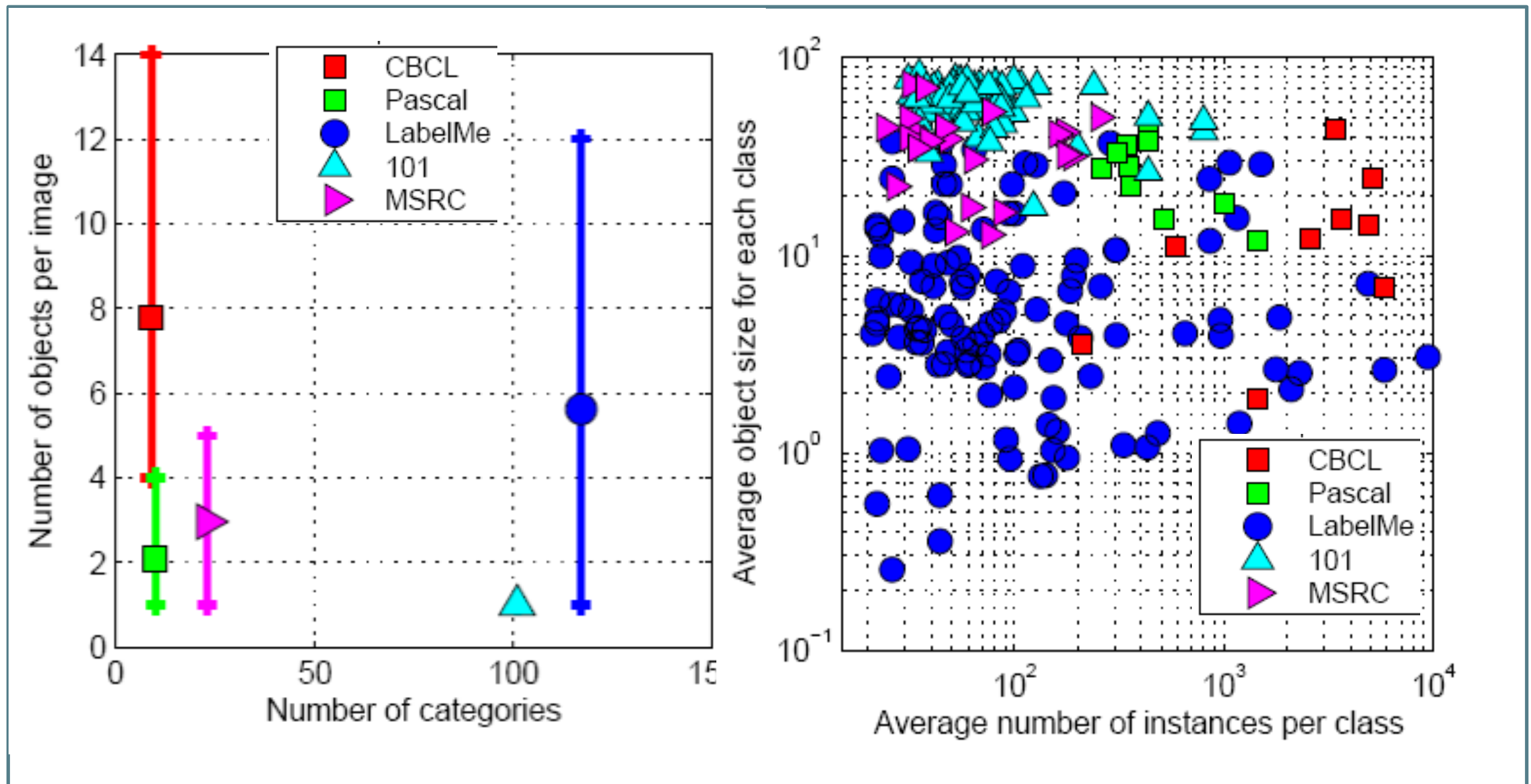


Figure 2. Peekaboom. Peek tries to guess the word associated with an image slowly revealed by "Boorn."

Is my dataset good?

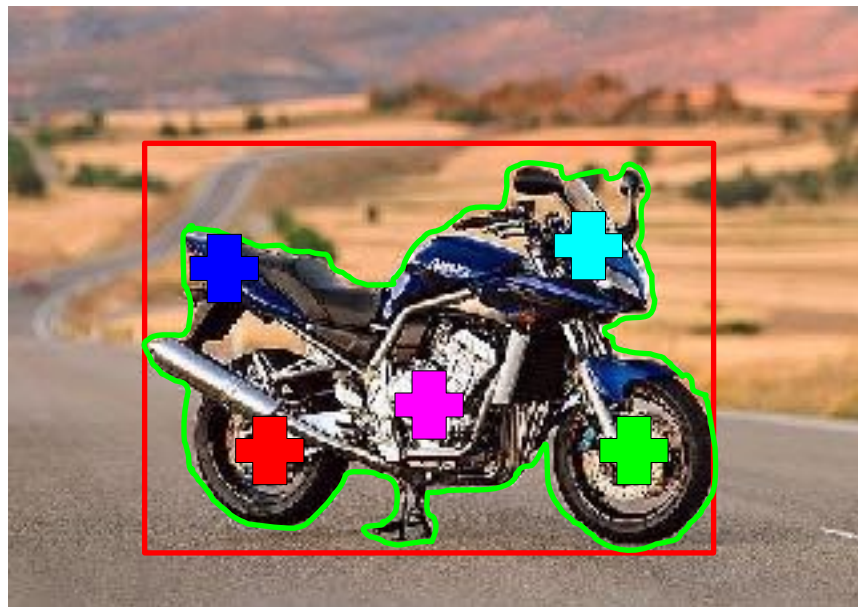
- Enough classes? Enough examples? Enough variability?
Good labels?



How to label training data?

- Depends on the application/problem

Contains a motorbike

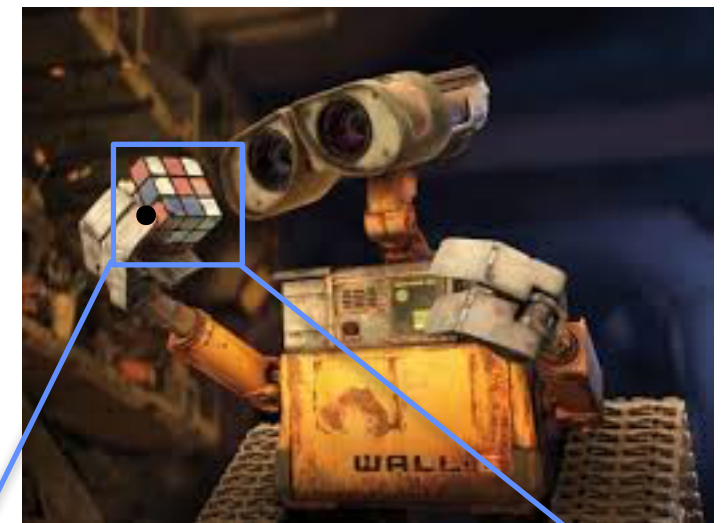


ML pipeline & DL

- **Collect** data (labeled?), choose how to **represent** it?

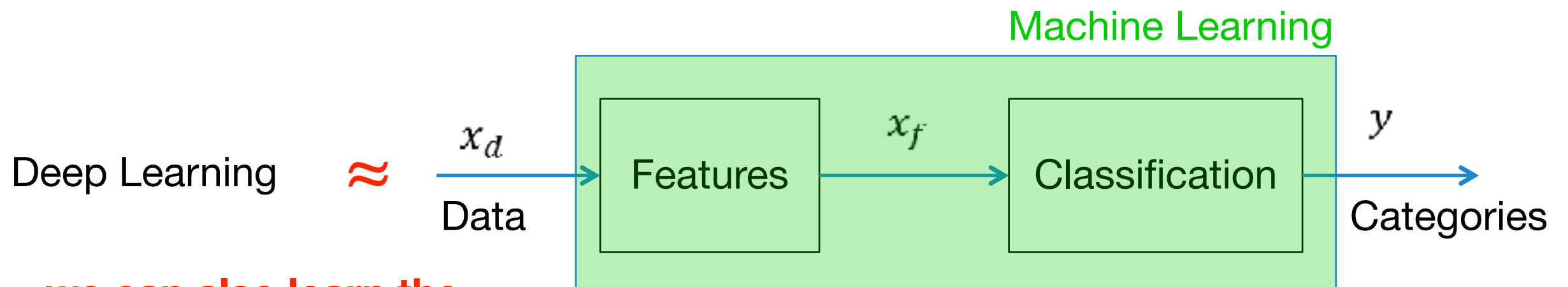
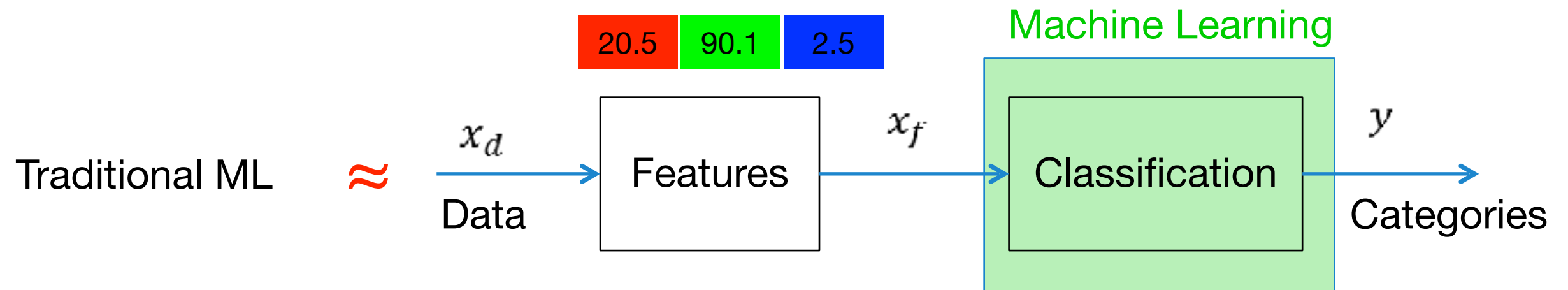
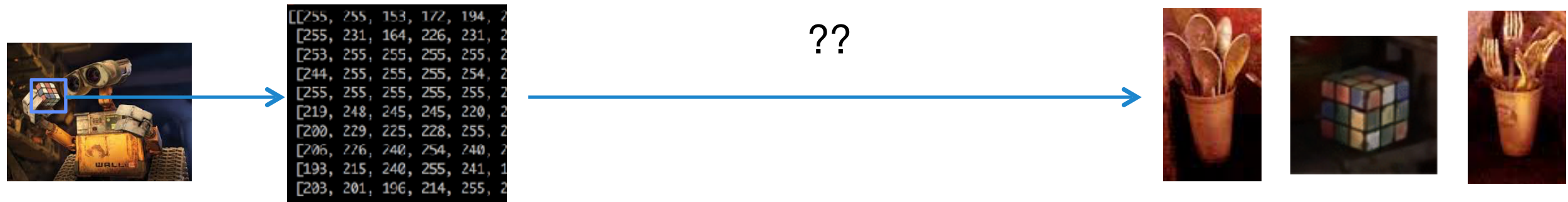
features: **VERY complicated and important *design* decision!!**

- **Learn**: organize the sample data (training data): fit/learn a model
- **Classify: evaluate** new data with regard to the model learnt (classify/recognize new content)



```
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 [203, 201, 196, 214, 255, 210, 180, 176, 254, 249, 243, 216]]
```


ML pipeline & **DL** pipeline



we can also learn the representations (features)

Next

- What's deep learning?
 - Related topics
 - DL pipeline
- **Fundamentals of DL**
 - **Review basic concepts**
 - NN and DNN

Bibliography - Resources for some of the materials today

- Stanford classes on deep learning for Computer Vision (<http://cs231n.stanford.edu>)
- Ian Goodfellow, Yoshua Bengio, Aaron Courville. ***Deep Learning***. MIT Press, 2016. <http://www.deeplearningbook.org>
- Book: ***Computer Vision: Algorithms and Applications*** (<http://szeliski.org/Book/>). Richard Szeliski, Steve Seitz.
Computer vision course slides. CSE 576 (Graduate Computer Vision) University of Washington (<http://courses.cs.washington.edu/courses/cse576/08sp/>)
- Fei-Fei, Fergus, Torralba: short course “recognising objects” (<http://people.csail.mit.edu/torralba/shortCourseRLOC>)